

Charlie Collins

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EDUCATION

Florida Institute of Technology, Melbourne, Florida

May 2025

Bachelor of Science, Computer Science

- 4.0 GPA
- Secretary of CPFIT - Florida Tech's Competitive Programming Club

SKILLS

Java, Python, C++, JavaScript, React, Node.js, HTML/CSS, INTEGRITY OS, MULTI debugger, Git, Subversion, MongoDB, SQL, Polaris, Jira, RTC, MASM, Jupyter, VSCode, Fusion 360, Linux (Ubuntu)

WORK EXPERIENCE

Wabtec, Melbourne, Florida

August 2024 - Present

Embedded Software Engineering co-op

- Created certificate management functionality that interfaces with the MQTT protocol
- Worked within a software engineering team following an agile development cycle
- Implemented a TLS interface for an embedded system, enhancing product cyber compliance

Wabtec, Melbourne, Florida

Summer 2024

Embedded Software Engineering Intern

- Developed software that runs on the INTEGRITY Real-Time Operating System
- Built a webpage to display diagnostic information from an embedded system using HTML
- Worked with the Polaris software platform to perform Static Application Security Testing (SAST)
- Assisted with the onboarding of a new enterprise Dynamic Application Security Testing (DAST) tool

Florida Tech Biomedical Image Computation Lab, Melbourne, Florida

August 2023 - May 2024

Undergrad Research Assistant

- Wrote a research paper presented at the 2024 IEEE Medical Imaging Conference
- Assisted in developing new algorithms for dynamic medical image reconstruction
- Conducted a statistical analysis of tracer distributions in SPECT medical images

Wabtec, Melbourne, Florida

Summer 2023

Embedded Software Engineering Intern

- Created a diagnostic menu using embedded C++ for the LOCOTROL product to aid technicians
- Ported a diagnostic feature from the legacy system to the current generation architecture
- Cultivated debugging skills with the Greenhills MULTI tool suite for embedded systems

RELATED EXPERIENCE

International Collegiate Programming Contest SER

February 2023; March 2024

- Awarded 2nd place in Division 2 of the Southeast USA region
- Strengthened my critical problem-solving skills using programming languages and tools
- Gained valuable experience designing and applying algorithms and data structures

Florida Tech Hackathon

November 2020

- Designed and implemented hardware and software to solve a Covid-19 problem
- Presented my work to a panel of judges and awarded first place in the competition

HONORS & AWARDS

- Distinguished Scholar Scholarship
- Dean's list